

Math 241, F04
Quiz 6

Name:
SSN: xxx-xx-

Instructions: There are totally 3 problems. You must show all of your work to receive a credit for a correct answer. No graphical calculator is allowed in the quiz and exam.

Problem 1. (6 points) Given $f(x, y) = 2x^2 + xy - y^2$, $\mathbf{p} = (3, -2)$ and $\mathbf{a} = \mathbf{i} - \mathbf{j}$, find the directional derivative of f at the point $\mathbf{p}$ in the direction of $\mathbf{a}$.

Solution: The unit vector $\mathbf{u}$ in the direction of $\mathbf{a}$ is:

$$\mathbf{u} = \frac{\mathbf{a}}{|\mathbf{a}|} = \frac{\mathbf{i} - \mathbf{j}}{\sqrt{2}} = \frac{1}{\sqrt{2}}\mathbf{i} - \frac{1}{\sqrt{2}}\mathbf{j}$$

We also know

$$f_x(x, y) = 4x + y, \quad f_y(x, y) = x - 2y,$$

thus,

$$f_x(3, -2) = 10, \quad f_y(3, -2) = 7.$$

Consequently, we have

$$D_{\mathbf{u}}f(3, -2) = 10 \cdot \frac{1}{\sqrt{2}} + 7 \cdot \left(-\frac{1}{\sqrt{2}}\right) = \frac{3}{\sqrt{2}}$$

Problem 2. (7 points) $w = e^{x^2+y^2}$, $x = s + t$, $y = s - t$, find $\frac{\partial w}{\partial t}$ using chain rule and express the final answer in term of s and t .

Solution:

$$\begin{aligned} \frac{\partial w}{\partial x} &= 2xe^{x^2+y^2}, & \frac{\partial w}{\partial y} &= 2ye^{x^2+y^2}, \\ \frac{\partial x}{\partial t} &= 1, & \frac{\partial y}{\partial t} &= -1 \end{aligned}$$

So we have

$$\begin{aligned} \frac{\partial w}{\partial t} &= \frac{\partial w}{\partial x} \frac{\partial x}{\partial t} + \frac{\partial w}{\partial y} \frac{\partial y}{\partial t} \\ &= 2xe^{x^2+y^2} \cdot 1 + 2ye^{x^2+y^2} \cdot (-1) \\ &= 2e^{x^2+y^2}(x - y) \\ &= 2e^{(s+t)^2+(s-t)^2}((s+t) - (s-t)) \\ &= 4te^{2(s^2+t^2)} \end{aligned}$$

Problem 3. (7 points) Given a surface determined by $x^2 + y^2 - z^2 = 4$, find the equation of the tangent plane to this surface at the point $(2, 1, 1)$.

Solution: Let $F(x, y, z) = x^2 + y^2 - z^2$, then

$$F_x(x, y, z) = 2x, \quad F_y(x, y, z) = 2y, \quad F_z(x, y, z) = -2z,$$

then

$$F_x(2, 1, 1) = 4, \quad F_y(2, 1, 1) = 2, \quad F_z(2, 1, 1) = -2.$$

So the equation of the tangent plane at $(2, 1, 1)$ is

$$\begin{aligned} 4(x - 2) + 2(y - 1) - 2(z - 1) &= 0 \\ 4x + 2y - 2z &= 8 \\ 2x + y - z &= 4 \end{aligned}$$